

Quantum Mechanics III

Lecture 6

Identical Particles and Count Correlations

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$[a_i, a_j^\dagger]_{\mp} = \delta_{ij}$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$g^{(2)}(\tau) = \langle : n(t)n(t + \tau) : \rangle / \langle n \rangle^2$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$P_{coin} = \frac{1}{2}(1 - |\langle \psi_1 | \psi_2 \rangle|^2)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

Identical Particles and Count Correlations

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

Reading: Park, ch. 6; Identical-Particle Registry IP-8; QDC coincidence block